Skills Summary

Reading Line Plots Marked in Fractions

Use this reference while completing your worksheet or when studying.

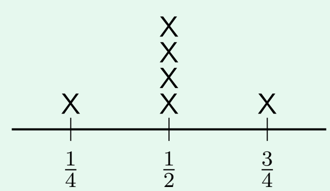
Skill 1 — Reading a plot marked in fractions

Two kinds of number live on one plot. Every X stands for one thing that was measured. The fraction printed underneath says how big that one thing was.

So *how many* questions are answered by counting X's, and the answer is a whole number. *How much* questions are answered by the fractions, and the answer is a length or an amount.

Check the scale before you read a single stack: one plot may count by eighths and the next by quarters, even though the marks look identical.

Example: How many ribbons were $\frac{1}{2}$ inch long?



Ribbon length (inches)

Step 1. The question asks how many *ribbons*, so the answer is the height of a stack — find the $\frac{1}{2}$ mark and count up, do not read the fraction.
Step 2. Above the middle mark: X, X, X, X.
$\Rightarrow$ **4** ribbons were half an inch long.

Try it: On the same plot, how many ribbons were $\frac{3}{4}$ inch long?

check: **1** ribbon

Skill 2 — Comparing what the plot shows

Decide first what kind of answer the question wants.

How many more of one than the other? → subtract the two stack heights, and the answer is a count.

How much longer / how much more? → subtract the two fractions from the bottom of the line, and the answer is a length or an amount. The longest is the rightmost X; the shortest is the leftmost.

Example: On the ribbon plot, how much longer is the longest ribbon than the shortest one?
Step 1. "How much longer" asks about size, so this one uses the fractions along the bottom, not the stacks.
Step 2. Rightmost X is on $\frac{3}{4}$, leftmost is on $\frac{1}{4}$, and $\frac{3}{4} - \frac{1}{4} = \frac{2}{4}$.
⇒ $\mathbf{\frac{1}{2}}$ of an inch longer.

Try it: On the same plot, how many *more* ribbons were $\frac{1}{2}$ inch long than were $\frac{1}{4}$ inch long?

check: **3** more ribbons

Skill 3 — Totalling and sharing equally

To total: write down one fraction for every X — a stack of four $\frac{1}{2}$'s means four halves, not one — rewrite them all over the same denominator, and add.

To share equally: divide that total by the number of X's (not by the number of marks on the line). The share is what every item *would* be if they were all levelled out, so it need not be a size anything actually was.

Example: On the ribbon plot, how much ribbon is there altogether, and how long would each ribbon be if they were all made the same?
Step 1. Six X's means six lengths to add, so rewrite everything in quarters — the smallest unit on this line.
Step 2. $\frac{1}{4} + \frac{2}{4} + \frac{2}{4} + \frac{2}{4} + \frac{2}{4} + \frac{3}{4} = \frac{12}{4}$ inches, and sharing that among six ribbons gives $12 \div 6 = 2$ quarters each.
⇒ **3** inches in total, or $\mathbf{\frac{1}{2}}$ of an inch each.

Try it: Four ribbons measure $\frac{1}{2}$, $\frac{3}{4}$, $\frac{3}{4}$ and 1 inch. Made all the same, how long would each be?

check: $\mathbf{\frac{3}{4}}$ of an inch each